Lecture 03.13: Sum of N Numbers

📌 Executive Summary

Sum of first N natural numbers: Iterative O(N) accumulation compared against Gauss's closed-form O(1) formula.

🔬 Theoretical Foundations & Principles

1. Loop Accumulation ($mathcalO(N)$)

Summing elements iteratively executes N additions and pointer rebinding operations:

$$S_N = \sum_{i=1}^{N} i$$

2. Gauss's Closed-Form Equation ($mathcalO(1)$)

Using pairing symmetries, Carl Friedrich Gauss proved the closed-form arithmetic sum:

$$S_N = \frac{N(N+1)}{2}$$

Executing in $\mathcal{O}(1)$ constant time and zero auxiliary space.

💻 Production Implementation & Code

```
def sum_iterative(n: int) -> int:
    total = 0
    for i in range(1, n + 1):
        total += i
    return total
```

```
def sum_gauss(n: int) -> int:
    return (n * (n + 1)) // 2

n_val = 1_000_000
print("Loop Sum:", sum_iterative(n_val))
print("Gauss Sum:", sum_gauss(n_val))
```

⚙️ Algorithmic Complexity & Systems Analysis

- **Time Complexity:** Detailed in respective algorithmic operations.
- **Auxiliary Space:** Memory bounds and CPython frame management.